

CSI4901- Capstone Project

INTELLIGENT FEEDBACK SYSTEM FOR ENHANCING PLACEMENT EXPERIENCES

Submitted in partial fulfilment of the requirements for the degree of

Master of Technology [Integrated]

in

**Computer Science and Engineering
(Data Science)**

by

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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

April, 2026

DECLARATION

I hereby declare that the project entitled “INTELLIGENT FEEDBACK SYSTEM FOR ENHANCING PLACEMENT EXPERIENCES” submitted by me, for the award of the degree of Master of Technology [Integrated] in Computer Science and Engineering (Data Science) to VIT is a record of bonafide work carried out by me under the supervision of Prof. Kanagaraj K, School of Computer Science and Engineering.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

JAISHWANTH A (21MID0149)

Date :

CERTIFICATE

This is to certify that the project entitled “INTELLIGENT FEEDBACK SYSTEM FOR ENHANCING PLACEMENT EXPERIENCES” submitted by JAISHWANTH A (21MID0149), School of Computer Science and Engineering, VIT, for the award of the degree of Master of Technology [Integrated] in Computer Science and Engineering (Data Science), is a record of bonafide work carried out by the student under my super- vision during academic year 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place: Vellore

Signature of the Guide

Date :

Internal Examiner

External Examiner

Head of the Department

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EXECUTIVE SUMMARY

Accumulated data on student course feedback and placements/internships is valuable to evaluate the quality of our education, but analysing the course feedback provided by students through text is a manual process that takes a lot of time, is subjective, and does not scale well when students provide a large amount of feedback.

This project will develop an intelligent system which will provide real-time, automated sentiment analyses for both student feedback and placement related using classical machine learning techniques and hybrid approaches. The system will preprocess unstructured textual feedback using commonly accepted text normalisation techniques; then, the textual feedback features will be converted into a numeric format using TF-IDF feature extraction. The resulting feedback will be classified using a variety of machine learning classifiers, such as SVM, random forest, KNN, and logistic regression, in order to classify feedback into sentiment categories.

In order to improve the system's robustness and performance, ensemble learning techniques will also be used to compare various classification models based on standard metrics like accuracy, precision, recall, and F1 score. The feedback will also be grouped for analysis purposes based on the aspects of location, size of the company or job role.

Additionally, this project aims to support organisations while respecting their privacy obligations using ethical data methods. The system proposed will help educational institutions to analyse feedback from students systematically to identify possible improvements and support data based decisions when improving the quality of education and placement success.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
HR	Human Resources
KNN	K-Nearest Neighbors
LIME	Local Interpretable Model Agnostic Explanations
ML	Machine Learning
NLP	Natural Language Processing
SGD	Stochastic Gradient Descent
SHAP	Shapley Additive Explanations
SVM	Support Vector Machine
TF-IDF	Term Frequency Inverse Document Frequency
XAI	Explainable Artificial Intelligence

SYMBOLS AND NOTATIONS

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Organisations and students continue to rely more than ever on user generated text data, such as employee reviews, feedback forms, and online ratings, to judge workforce satisfaction and culture. Various sites, such as AmbitionBox, produce hundreds of thousands of employee reviews each day that express opinions as to how an employee feels about their job description, the practices of their superiors, how well they balance work and home life, and their growth potential.

Manual analysis of hundreds of thousands of pages of unstructured text data is highly inefficient and time consuming because it is prone to human bias. Different reviewers may interpret the same review with different meanings and come to significantly different conclusions regarding overall impressions. As organisations grow, manual methods of analysis become impractical.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have made it possible to establish automated sentiment analysis systems capable of efficiently processing large volumes of text. The information retrieved from these texts can be categorized into sentiment expressions like positive or negative through these automated sentiment analysis systems and can therefore be used to quickly assess overall perceptions.

The goal of this project is to create an employee review sentiment analysis system that utilises the AI, ML, and NLP advancements referred to above, while ensuring that it is scalable, accurate and has interpreted results for employee reviews in the Indian job market.

1.2 MOTIVATION

This project was initiated because organisations and students are becoming more dependent on online feedback from employees when recruiting or making organisational decisions, and therefore use online employee reviews in their recruitment processes. Job seekers often rely on these online employee reviews to judge an organisation's fit

as a future employer. Similarly, organisations and students also rely on reviews to make changes to policies, procedures, and the overall workplace culture.

Manual sentiment analysis suffers from many limitations, including the following:

- Subjective interpretations of reviews made by human analysts.
- Inability to scale up to large datasets.
- Inconsistent and repeatable results from manual sentiment analysis.
- No transparency in the decision-making process.

Many AI systems also suffer from being a 'black box' where the AI system will produce a prediction without clarity on why it received a particular sentiment. This is the reason for the low level of trust in these automated systems. The motivation for this project is:

- The need for an automated, unbiased approach to producing sentiment analysis.
- The need for AI systems that are both transparent and provide us with the capability to explain how they reached the results produced.
- There is a significant upside in using ML and NLP to analyse real-world datasets.
- There is a growing demand for decision support tools such as recruitment and Human Resources (HR) analytics.

1.3 SCOPE OF THE PROJECT

The project will address offline sentiment analysis for employees through a dataset that is a collection of employee reviews made from a structured data source. For this project, a dataset from AmbitionBox has been used, but any structured employee review dataset can be used.

Within the Scope:

- Text preprocessing and data clean up
- Term Frequency Inverse Document Frequency (TF-IDF) feature extraction
- Train multiple machine learning models

- Compare and evaluate the models
- Use of ensemble learning techniques
- Use of Shapley Additive Explanations (SHAP) and Local Interpretable Model Agnostic Explanations (LIME) for explainable AI analysis
- Aggregate the sentiment data across companies, sectors, and locations

Outside the Scope:

- Multilingual sentiment analysis

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

For the literature review, I have taken 50 papers from journals and conferences that have happened in the recent past.

No.	Author(s), Year	Title	Domain/Data	Techniques	Key Contributions	Limitations
1	IEEE, 2019 IEEE et al. (2019b)	Aspect Based Opinion Min- ing on Student Feedback	Student feed- back	LSTM, ABSA	Fine grained sentiment for faculty evaluation	Focus only on teaching
2	IEEE, 2022 IEEE et al. (2022b)	Educational DSS using Sentiment Analysis	Student feed- back	NLP	Decision support	No multilingual handling
3	IEEE, 2021 IEEE et al. (2021e)	Sentiment Anal- ysis in MOOCs Survey	MOOCs	Survey	Overview of methods	No implementation
4	IEEE, 2020 IEEE et al. (2020a)	Analysis of Stu- dent Essays using NLP	Essays	NLP	Shows NLP usefulness	No sentiment polarity
5	IEEE, 2016 IEEE et al. (2016a)	Feature Level Opinion Mining	Education feedback	Rule mining	Interpretable extraction	Needs structured data
6	IEEE, 2019 IEEE et al. (2019a)	ABSA on Student Feedback	Faculty feed- back	Deep learn- ing	Better interpretability	Ignores placement
7	IEEE, 2008 Pang and Lee (2008)	Opinion Mining	Reviews	NB, SVM	Foundational methods	Coarse sentiment
8	IEEE Global, 2018 IEEE et al. (2018b)	Sentiment in MOOCs	MOOC reviews	ML	Trend detection	No ABSA
9	IEEE, 2017 IEEE et al. (2017)	Internship Feed- back Analysis	Internship	NLP	Placement relevance	Small dataset
10	IEEE, 2022 IEEE et al. (2022c)	Educational DSS Extended	Institutions	NLP	Real time analytics	Limited ML comparison
11	IEEE Ac- cess, 2020 IEEE et al. (2020d)	Semantic Net- works	Student feed- back	Semantic nets	Improves accuracy	High complexity

No.	Author(s), Year	Title	Domain/Data	Techniques	Key Contributions	Limitations
12	ResearchGate, 2017 ResearchGate et al. (2017)	Opinion Mining Student Feedback	Education	ML	ML better than lexicon	No ABSA
13	IEEE Access, 2023 IEEE et al. (2023i)	Teaching Evaluation Sentiment	Course reviews	ML, DL	DL improves accuracy	Not placement based
14	IEEE, 2016 IEEE et al. (2016b)	Hybrid Sentiment Models	Social Edu	Hybrid	Better interpretability	Manual lexicon
15	IEEE, 2022 IEEE et al. (2022i)	Code Mixed Sentiment	Mixed text	Multilingual NLP	Handles mixed language	Not education specific
16	IEEE, 2023 IEEE et al. (2023c)	Ensemble Learning	Text	RF, Voting	Higher accuracy	High cost
17	IEEE, 2024 IEEE et al. (2024c)	Explainable Sentiment Survey	NLP	XAI	Improves trust	Theoretical
18	IEEE, 2022 IEEE et al. (2022f)	Resume Classification	Resume	NLP	Job mapping	No sentiment
19	EduRABSA EduRABSA et al. (2023)	Education Dataset	Reviews	ABSA	Benchmark dataset	No placement
20	IEEE, 2023 IEEE et al. (2023a)	Automated Feedback System	Institutional	DistilBERT	End to end system	Expensive
21	IEEE, 2021 IEEE et al. (2021d)	MOOC Sentiment DL	MOOCs	CNN	Learner insights	Needs large data
22	IEEE, 2020 IEEE et al. (2020e)	TF IDF vs Embeddings	Interviews	NLP	Semantic understanding	Limited generalisation
23	IEEE, 2022 IEEE et al. (2022j)	SVM and RF Sentiment	Reviews	SVM, RF	Robust performance	No ABSA
24	IEEE Access, 2024 IEEE et al. (2024b)	XAI Sentiment	Reviews	SHAP	Transparency	Complexity

No.	Author(s), Year	Title	Domain/Data	Techniques	Key Contributions	Limitations
25	IEEE, 2023 IEEE et al. (2023b)	DL Sentiment Student	Course feed- back	CNN, LSTM	High accuracy	Costly
26	IEEE, 2019 IEEE et al. (2019c)	NLP Teaching Enhancement	Education	NLP	Learning analytics	No benchmarking
27	IEEE, 2021 IEEE et al. (2021c)	Emotion Aware MOOCs	MOOCs	Emotion models	Better accuracy	Hard annotation
28	IEEE, 2018 IEEE et al. (2018a)	Imbalanced Met- rics	Data	Metrics	Evaluation guidance	No case study
29	IEEE, 2020 IEEE et al. (2020c)	Hybrid Sentiment Models	Movie reviews	Hybrid	Better accuracy	Domain mismatch
30	IEEE, 2022 IEEE et al. (2022g)	Resume via LinkedIn	Job data	NLP	Placement mapping	No sentiment
31	Springer, 2024 Springer et al. (2024)	Text Mining Con- cepts	Text analyt- ics	Mining	Fundamentals	No experiments
32	ResearchGate, 2023 Re- searchGate et al. (2023)	Academic vs Job Performance	Student data	Stats	Links academics	No NLP
33	ArXiv, 2025 ArXiv et al. (2025)	EduRABSA Dataset	Reviews	ABSA	Fine grained data	Limited deployment
34	IJIRTE, 2019 IJIRTE et al. (2019)	Student Senti- ment ML	Education	ML	Validates ML	Small data
35	IEEE, 2023 IEEE et al. (2023h)	Multilingual Transformers	Multilingual	mBERT	Indian languages	Latency
36	IEEE, 2022 IEEE et al. (2022e)	Feature Selection	Student data	Feature engg.	Improves accuracy	Manual tuning
37	IEEE, 2023 IEEE et al. (2023e)	Job Review Sen- timent	Job reviews	NLP	Placement relevance	Not education specific

No.	Author(s), Year	Title	Domain/Data	Techniques	Key Contributions	Limitations
38	IEEE, 2021 IEEE et al. (2021b)	Domain Adapta- tion	Job NLP	Transfer learning	Cross domain	Needs data
39	IEEE, 2020 IEEE et al. (2020b)	Data Cleaning NLP	Education	Preprocessing	Better performance	Dataset specific
40	IEEE, 2022 IEEE et al. (2022a)	Imbalance Han- dling	Text	SMOTE	Better recall	Overfitting risk
41	IEEE, 2023 IEEE et al. (2023k)	ML Visualisation	Analytics	Dashboards	Better decisions	No sentiment
42	IEEE, 2024 IEEE et al. (2024d)	Multilingual Education Senti- ment	Student feed- back	NLP	Wider coverage	Weak ABSA
43	IEEE, 2023 IEEE et al. (2023f)	Job Role Classifi- cation	Job feedback	NLP	Placement support	No sentiment
44	IEEE, 2022 IEEE et al. (2022d)	Ethics and Pri- vacy in Education Data	Student data	Privacy	Compliance	No ML
45	IEEE, 2021 IEEE et al. (2021a)	ABSA Short Text	Reviews	ABSA	Sparse handling	Ambiguity
46	IEEE, 2023 IEEE et al. (2023j)	Transformer vs ML	Text	ML vs DL	Trade off study	No placement
47	IEEE, 2022 IEEE et al. (2022h)	HR Sentiment	Employee reviews	NLP	Placement adaptable	Not student focused
48	IEEE, 2023 IEEE et al. (2023g)	LLM Sentiment Survey	NLP	Prompting	Future direction	Heavy models
49	IEEE, 2024 IEEE et al. (2024a)	Explainability in Education NLP	Education	XAI	Trust building	Overhead
50	IEEE, 2023 IEEE et al. (2023d)	Integrated Feed- back System	Education and place- ment	NLP ML	Automation	Limited multilingual

2.2 RESEARCH GAP

Existing methodologies contain these issues:

- Manual sentiment analysis is both time consuming and subjective, therefore, inconsistent.
- A one model solution does not offer good reliability or consistency.
- Conventionally based machine learning algorithms do not provide sufficient clarity around their results.
- Processing large volumes of text data has high time complexity.
- Lack of clarity in what features affect decision making
- No aggregation of data across multiple companies, industries, and geographic locations.
- Combining and interpreting noisy and informal employee comments can be difficult.

2.3 OBJECTIVES

The primary objective of this project is to implement an accurate, easily expandable and interpretable AI based sentiment analysis system.

- Data cleaning and preprocessing of employee review data. Transformation of textual information to numeric form with TF-IDF.
- Train 6 machine learning classification models using ensemble modelling to increase accuracy.
- Using Explainable AI methods for transparency.
- Creation of groups such as location, sector, and size of the company to let the user decide according to their needs.

2.4 PROBLEM STATEMENT

Employee feedback data is often in text format, making it challenging for organisations and students to obtain valuable insights due to the large volume of this unstructured data. Existing approaches are either reliant upon manual input, which is subjective in nature or just like a blackbox simply do not provide any type of explanation as to why a particular sentiment was predicted. Therefore, there is a need for an automated system that accurately predicts sentiment but also explains the reasoning behind such predictions.

2.5 PROJECT PLAN

Employee feedback data is often in text format, making it challenging for organisations and students to obtain valuable insights due to the large volume of this unstructured data. Existing approaches are either reliant upon manual input, which is subjective in nature or just like a blackbox simply do not provide any type of explanation as to why a particular sentiment was predicted. Therefore, there is a need for an automated system that accurately predicts sentiment but also explains the reasoning behind such predictions.

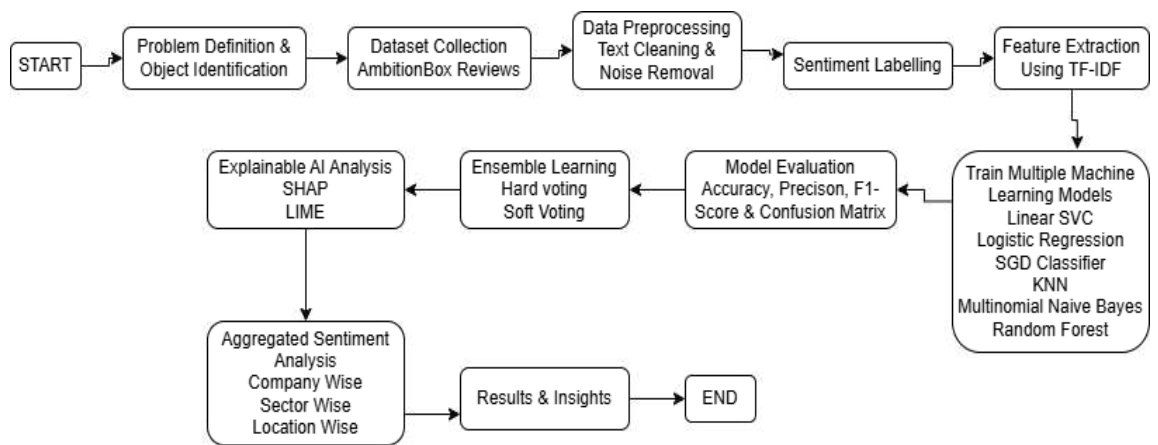


Figure 2.1: Project Plan Flow Chart

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

- The system will accept employee reviews as input
- The system will preprocess and clean text data
- The system will classify sentiment according to the set parameters
- The system will train multiple ML models
- The system will evaluate models using performance measures
- The system will provide explainable predictions with Explainable Artificial Intelligence (XAI)
- The system will generate grouped sentiment reports

3.1.2 Non Functional Requirements

- The system must be reliable and consistent
- The system must handle large datasets efficiently
- The system must ensure a reasonable execution time
- The system must be easy to interpret

3.2 FEASIBILITY STUDY

A feasibility study was conducted to evaluate whether the project is practical, cost effective and socially acceptable. This analysis ensures that the proposed solution can be developed with the given conditions.

3.2.1 Technical Feasibility

The project is technically feasible as it is developed using established and widely used technologies in the field of Machine Learning and Natural Language Processing. The system is implemented using Python and open source libraries such as Scikit-learn, Pandas, NumPy, and SHAP, which are reliable, efficient.

The algorithms used in this project, include Linear SVC, Logistic Regression, Naive Bayes, KNN, Random Forest, and ensemble methods. These models are efficient and suitable for medium to large scale datasets such as the one used in this work.

The use of TF-IDF for feature extraction ensures compatibility with classical ML models and enables explainability.

3.2.2 Economic Feasibility

The project is economically feasible as it does not require any financial investment. All the tools, libraries, and development environments used in this project are open source and freely available to use. There is no requirement for licensed or paid software or libraries.

This can be executed on any PC with the hardware and software specifications that are mentioned. This allows it to be implemented on any PC easily.

3.2.3 Social Feasibility

Socially, the project is very feasible. The system assists job seekers, mainly students, to choose their career wisely by reading authentic employee reviews. This reduced their reliance on unreliable review that are openly available on the interweb. The use of explainable AI also helps improve explainability and interpretability.

It reduces reliance on traditional text based reviews that are found online in abundance. Explainable AI also give explanations making it transparent and trustable in comparison to existing systems.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

- Processor: Intel i5 10th gen or AMD Ryzen 5 or Apple M1 or above

- RAM: Minimum 8 GB
- Storage: Minimum 256 GB
- Architecture: 64 bit System

3.3.2 Software Specification

- Operating System: Windows 10/11, Linux, Mac OS
- Language: Python
- Libraries: NumPy, Pandas, Scikit Learn, Matplotlib, SHAP & LIME
- IDE: Jupyter Notebook, VS Code, Google Colab

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The system architecture shows the flow of employee review data from AmbitionBox, from data collection and preprocessing to sentiment classification, evaluation, explainable AI analysis, and aggregated insights generation. Each layer is responsible for performing a specific task or function to ensure the best results are obtained.

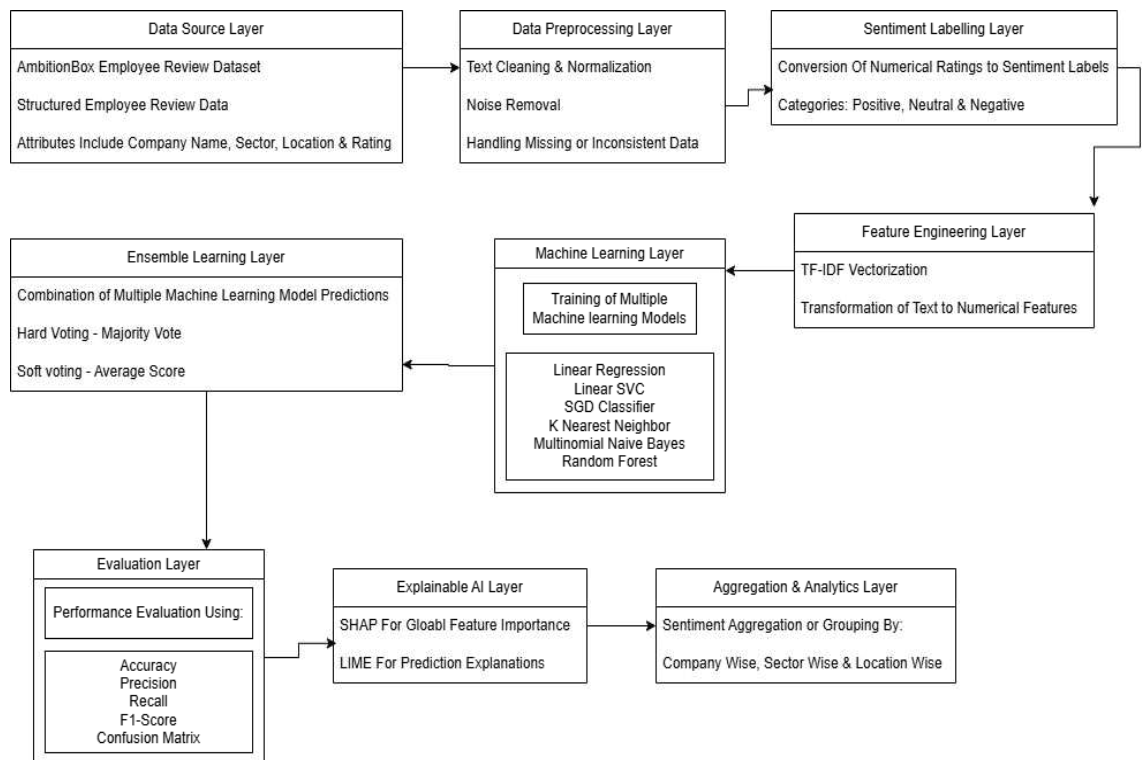


Figure 4.1: System Architecture Class Diagram

4.2 DESIGN

4.2.1 Data Flow Diagram

The data flow diagram shows the movement of the employee review data obtained from AmbitionBox through all the seven stages, starting from data collection to feature extraction and finally to sentiment aggregation.

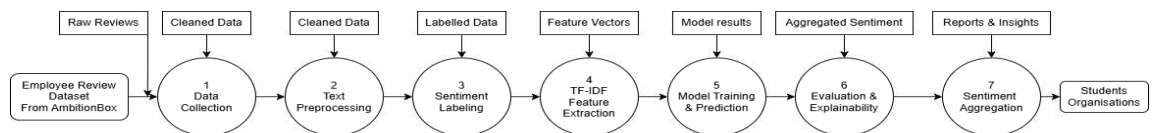


Figure 4.2: Data Flow Diagram

4.2.2 Class Diagram

The class diagram shows the logical structure of the system, this shows the interaction between all the various modules with their different attributes and methods.

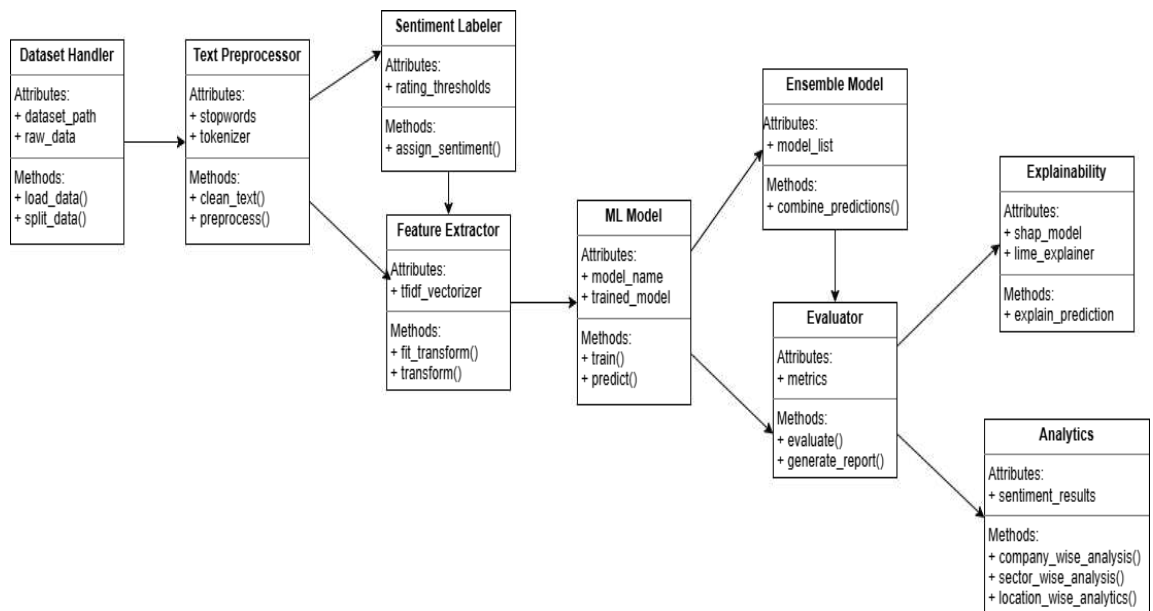


Figure 4.3: Class Diagram

CHAPTER 5

METHODOLOGY AND TESTING

5.1 DATA ACQUISITION

The dataset is sourced from AmbitionBox. It contains Indian companies across multiple sectors. Real employee generated reviews are present. This ensures authenticity and relevance. The dataset has a usability rating of 10.0 on kaggle.com.

5.2 TEXT PREPROCESSING

Text is cleaned by the removal of noise such as symbols numbers. Then case normalisation is done by converting all text to lower case. Missing and invalid data are handled. Finally, the creation of a cleaned text field that involves the removal of special characters, punctuation, URLs, and HTML tags.

5.3 SENTIMENT LABELLING

The ratings are converted to sentiment classes with the parameters defined by us. This enables supervised learning. Also reduces human bias by sorting the sentiments according to the parameters.

5.4 FEATURE ENGINEERING

TF-IDF is used to convert textual data to numeric vectors. The importance of words is captured by extracting the features.

5.5 MACHINE LEARNING MODELS

Nine Machine Learning models were trained to ensure diversity in linear, probabilistic, distance based, and ensemble learners. This also enables us to analyse the strengths and weaknesses of each approach.

5.6 ENSEMBLE LEARNING

Both the hard and soft voting strategies have been made use of to get the best outcome. This combines predictions from the multiple trained ML models. Improves robustness and generalisation.

5.7 EXPLAINABLE AI

SHAP and LIME explainable AI models are made use of so that the end user is aware as to why a decision has been taken. SHAP is used for global interpretations, it makes use of concepts from game theory and assigns weights to each feature. The feature with more weight is considered to be more important. LIME is made use of for local interpretable explanations. It explains the individual predictions.

5.8 EVALUATION

The different machine learning models are evaluated using various performance measures such as accuracy score, precision score, recall score, F1-score, and confusion matrix. Then ensemble learning methods are used in the form of hard voting and soft voting to get the best performance from each one of the models. Hard voting takes the majority of the decision, whereas soft voting takes the average of the decisions and gives you the result.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 IMPLEMENTATION

To create an interactive dashboard to support the python program we have made use of HTML, CSS and JS.

You can enter the name of your company and select the type of company it is. It has a few options such as private, public, startup and multinational company. Once you have done that, you can select the sector your company works in. Here as well, you can choose from the given options such as IT services and consulting, financial services, healthcare, manufacturing, and education. You will now have to give the location of the company or the particular location or branch you work for. This will help us make location based sentiment analysis. Once all of these details have been filled out, you will have to score your company on a scale of 1 to 5, similar to a star system. Finally, you have to give a small writeup about your experience.

Once you click on the "Analyse Sentiment" button on the top right it will give you the results based on the information you have entered.

6.2 IMPLEMENTATION IMAGES

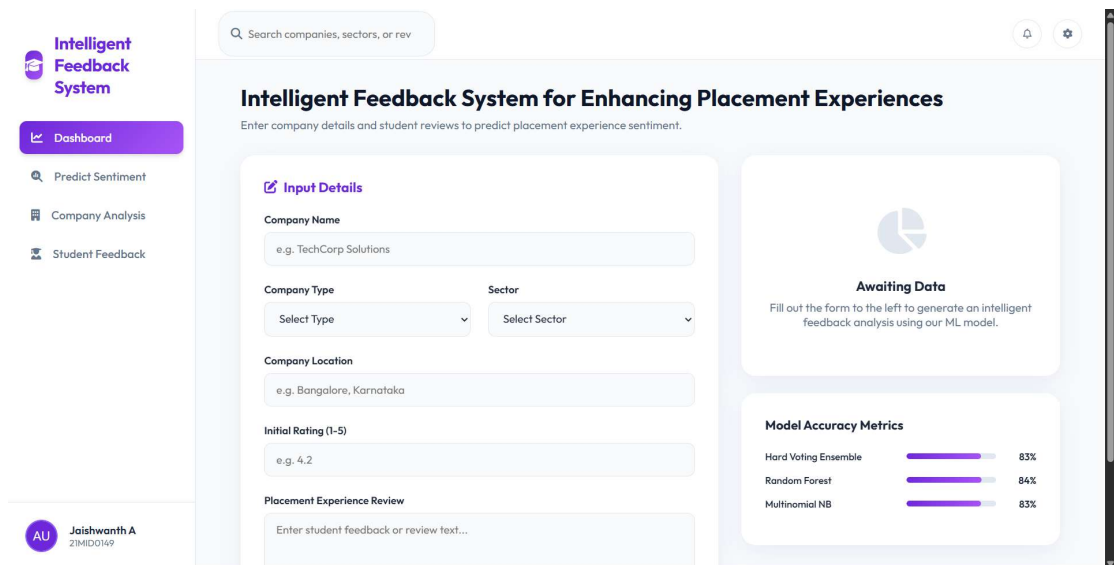


Figure 6.1: Home Page or Landing Page

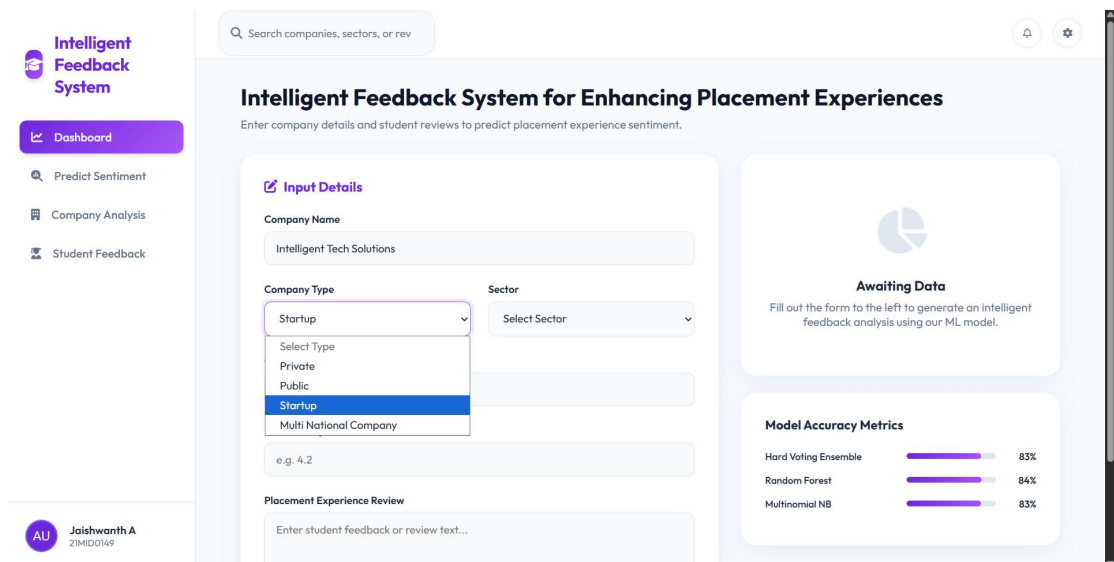


Figure 6.2: Entering Company Name and Company Type

Intelligent Feedback System

Search companies, sectors, or rev

Intelligent Feedback System for Enhancing Placement Experiences

Enter company details and student reviews to predict placement experience sentiment.

Input Details

Company Name: Intelligent Tech Solutions

Company Type: Startup

Company Location: e.g. Bangalore, Karnataka

Initial Rating (1-5): e.g. 4.2

Sector: IT Services & Consulting

Placement Experience Review: Enter student feedback or review text...

Awaiting Data

Fill out the form to the left to generate an intelligent feedback analysis using our ML model.

Model Accuracy Metrics

Model	Accuracy
Hard Voting Ensemble	83%
Random Forest	84%
Multinomial NB	83%

Figure 6.3: Selecting the Sector of the Company

Intelligent Feedback System

Search companies, sectors, or rev

Intelligent Feedback System for Enhancing Placement Experiences

Enter company details and student reviews to predict placement experience sentiment.

Input Details

Company Name: Intelligent Tech Solutions

Company Type: Startup

Company Location: Bangalore, Karnataka

Initial Rating (1-5): 3.8

Sector: IT Services & Consulting

Placement Experience Review: New company established in 2023. Good working experience overall

Analysis Results

Positive

Confidence Score

82%

Based on ensemble voting model (SVC, Random Forest, Gradient Boosting)

Key Insights

Highly positive indicators regarding work environment.

Model Accuracy Metrics

Model	Accuracy
Hard Voting Ensemble	83%
Random Forest	84%
Multinomial NB	83%

Analyze Sentiment

Figure 6.4: Location, Rating and Write Up

**Intelligent Feedback System**

Dashboard

Predict Sentiment

Company Analysis

Student Feedback

AU

Jaishwanth A

ZIMD00149

skoda

Home

Feedback

Analysis

Reports

Settings

Input Details

Company Name

FinTech Corp

Company Type

Private

Sector

Financial Services

Company Location

Mysore, Karnataka

Initial Rating (1-5)

2

Placement Experience Review

Terrible company with bad experience

Analyze Sentiment

Analysis Results

Negative

85%

Confidence Score

Based on ensemble voting model (SVC, Random Forest, Gradient Boosting)

Key Insights

Multiple negative sentiment flags detected.

Model Accuracy Metrics

Hard Voting Ensemble

83%

Random Forest

84%

Multinomial NB

83%

Figure 6.5: Negative Review Example

22

CHAPTER 7

RESULTS AND DISCUSSION

7.1 EXPERIMENTAL RESULTS

The system was evaluated on nine different machine learning models and then with ensemble learning as well, and the following results were obtained:

- Linear SVC (Support Vector Machine (SVM)) - 83.20
- SVC (SVM) - 83.20
- Logistic Regression - 82.75
- SGD Classifier (Stochastic Gradient Descent (SGD)) - 83.15
- Decision Tree - 80.20
- K Nearest Neighbors (K-Nearest Neighbors (KNN)) - 79.65
- Multinomial Naive Bayes - 83.10
- Random Forest - 84.00
- Gradient Boosting - 82.60
- Hard Voting - 82.95
- Soft Voting - 83.10

From the above results, we can see that random forest classifier has the highest accuracy with 84.00 and the lowest is KNN with 79.65.

7.2 CONFUSION MATRIX ANALYSIS

Confusion Matrix helps us understand how each model performs across each one of the different sentiment classes.

Key observation across all models is:

- Positive class dominates predictions.

- Negative class has fewer samples making it harder to detect.

In Random Forest:

- Positive samples are mostly correctly classified.
- Some neutral samples are misclassified as positive.
- Negative samples are misclassified due to low samples.

7.3 KEY OBSERVATIONS

- Ensemble models outperform individual models.
- Linear models perform very well for the TF-IDF features.
- Absence of many negative samples affects classification accuracy.
- Negative sentiment detection is bad due to fewer samples available in the dataset.

7.4 MODEL COMPARISON AND INTERPRETATION

In the figure 7.12 below, I have compared the different performance measures of all nine machine learning models with one another.

7.4.1 Random Forest

- Best performing model
- Handles high dimensional TF-IDF features well
- Captures non-linear patterns

7.4.2 Linear Models

SVC, Logistic Regression and SGD are the linear models that we have used in this project.

- Performed consistently around 83
- Suitable for text classification
- Fast and efficient

7.4.3 Naive Bayes

- Performed well despite simplicity
- Effective for word frequency based features

7.4.4 Decision tree

- Lower accuracy due to overfitting
- Sensitive to dataset imbalance

7.4.5 K-Nearest Neighbors

- Lowest performance
- Affected by high dimensional feature space
- Distance calculations become unreliable

7.4.6 Gradient Boosting

- Good performance, but slightly lower than Random Forest
- More sensitive to parameter tuning

7.4.7 Key Observations from ML Models

- Ensemble models outperform individual models
- Linear models perform well for TF-IDF features
- Dataset imbalance affects classification accuracy
- Negative sentiment detection is weak due to fewer samples

7.5 BERT MODEL

The BERT model gives us an output of [8, 3]. First, this shows us the size of the batch and the number of classes. The data is shown as positive, negative and neutral.

7.6 COMPANY, SECTOR AND LOCATION WISE SENTIMENT

In figures 7.14, 7.15 and 7.16 you will be able to see the overall interpretation of the companies, sectors and industries. There are three columns namely, they are positive, neutral and negative.

7.7 RESULTS OBTAINED

Model: Linear SVC

Accuracy: 0.8320

Confusion Matrix:

```
[[ 1  1  6]
 [ 1 66 271]
 [ 1 56 1597]]
```

Model: Linear SVC	precision	recall	f1-score	support
Negative	0.33	0.12	0.18	8
Neutral	0.54	0.20	0.29	338
Positive	0.85	0.97	0.91	1654
accuracy			0.83	2000
macro avg	0.57	0.43	0.46	2000
weighted avg	0.80	0.83	0.80	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.1: Linear SVC Model Performance

Model: SVC

Accuracy: 0.8320

Confusion Matrix:

```
[[ 0  0  8]
 [ 0 25 313]
 [ 0 15 1639]]
```

Model: SVC	precision	recall	f1-score	support
Negative	0.00	0.00	0.00	8
Neutral	0.62	0.07	0.13	338
Positive	0.84	0.99	0.91	1654
accuracy			0.83	2000
macro avg	0.49	0.35	0.35	2000
weighted avg	0.80	0.83	0.77	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.2: SVC Model Performance

Model: Logistic Regression

Accuracy: 0.8275

Confusion Matrix:

```
[[ 0  0  8]
 [ 0  9 329]
 [ 0  8 1646]]
```

Model: Logistic Regression	precision	recall	f1-score	support
Negative	0.00	0.00	0.00	8
Neutral	0.53	0.03	0.05	338
Positive	0.83	1.00	0.91	1654
accuracy			0.83	2000
macro avg	0.45	0.34	0.32	2000
weighted avg	0.78	0.83	0.76	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.3: Logistic Regression Performance

Model: SGD Classifier

Accuracy: 0.8315

Confusion Matrix:

```
[[ 0  0  8]
 [ 0 23 315]
 [ 0 14 1640]]
```

Model: SGD Classifier	precision	recall	f1-score	support
Negative	0.00	0.00	0.00	8
Neutral	0.62	0.07	0.12	338
Positive	0.84	0.99	0.91	1654
accuracy			0.83	2000
macro avg	0.49	0.35	0.34	2000
weighted avg	0.80	0.83	0.77	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.4: SGD Classifier Performance

Model: Decision Tree

Accuracy: 0.8020

Confusion Matrix:

```
[[ 1  2  5]
 [ 1 103 234]
 [ 4 150 1500]]
```

Model: Decision Tree	precision	recall	f1-score	support
Negative	0.17	0.12	0.14	8
Neutral	0.40	0.30	0.35	338
Positive	0.86	0.91	0.88	1654
accuracy			0.80	2000
macro avg	0.48	0.45	0.46	2000
weighted avg	0.78	0.80	0.79	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.5: Decision Tree Performance

Model: K Nearest Neighbors

Accuracy: 0.8055

Confusion Matrix:

```
[[ 0  0  8]
 [ 0 32 306]
 [ 0 75 1579]]
```

Model: K Nearest Neighbors

	precision	recall	f1-score	support
Negative	0.00	0.00	0.00	8
Neutral	0.30	0.09	0.14	338
Positive	0.83	0.95	0.89	1654
accuracy			0.81	2000
macro avg	0.38	0.35	0.34	2000
weighted avg	0.74	0.81	0.76	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.6: KNN Performance

Model: Multinomial Naive Bayes

Accuracy: 0.8310

Confusion Matrix:

```
[[ 0  0  8]
 [ 0 12 326]
 [ 0  4 1650]]
```

Model: Multinomial Naive Bayes

	precision	recall	f1-score	support
Negative	0.00	0.00	0.00	8
Neutral	0.75	0.04	0.07	338
Positive	0.83	1.00	0.91	1654
accuracy			0.83	2000
macro avg	0.53	0.34	0.32	2000
weighted avg	0.81	0.83	0.76	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.7: Multinomial Naive Bayes Performance

Model: Random Forest

Accuracy: 0.8400

Confusion Matrix:

```
[[ 1  0  7]
 [ 0 79 259]
 [ 2 52 1600]]
```

Model: Random Forest	precision	recall	f1-score	support
Negative	0.33	0.12	0.18	8
Neutral	0.60	0.23	0.34	338
Positive	0.86	0.97	0.91	1654
accuracy			0.84	2000
macro avg	0.60	0.44	0.48	2000
weighted avg	0.81	0.84	0.81	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.8: Random Forest Performance

Model: Gradient Boosting

Accuracy: 0.8260

Confusion Matrix:

```
[[ 1  0  7]
 [ 0  4 334]
 [ 4  3 1647]]
```

Model: Gradient Boosting	precision	recall	f1-score	support
Negative	0.20	0.12	0.15	8
Neutral	0.57	0.01	0.02	338
Positive	0.83	1.00	0.90	1654
accuracy			0.83	2000
macro avg	0.53	0.38	0.36	2000
weighted avg	0.78	0.83	0.75	2000

(a) Accuracy & Confusion Matrix

(b) Classification Report

Figure 7.9: Gradient Boosting Performance

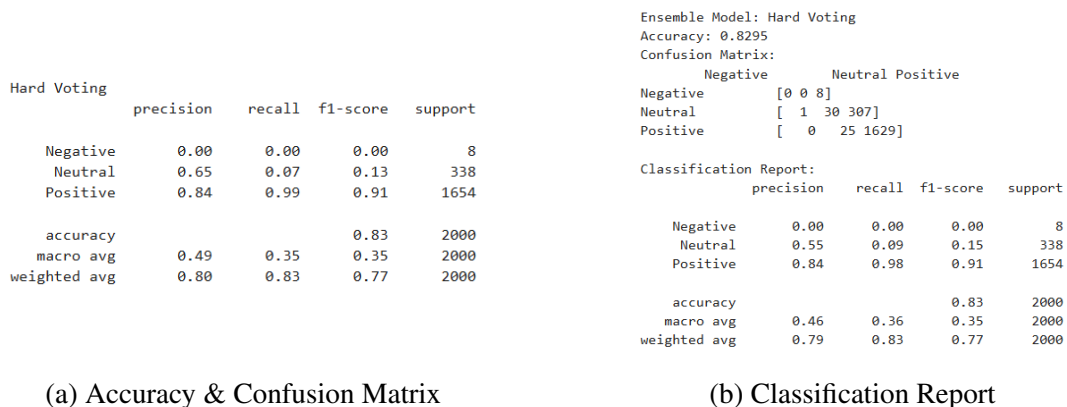


Figure 7.10: Hard Voting Ensemble

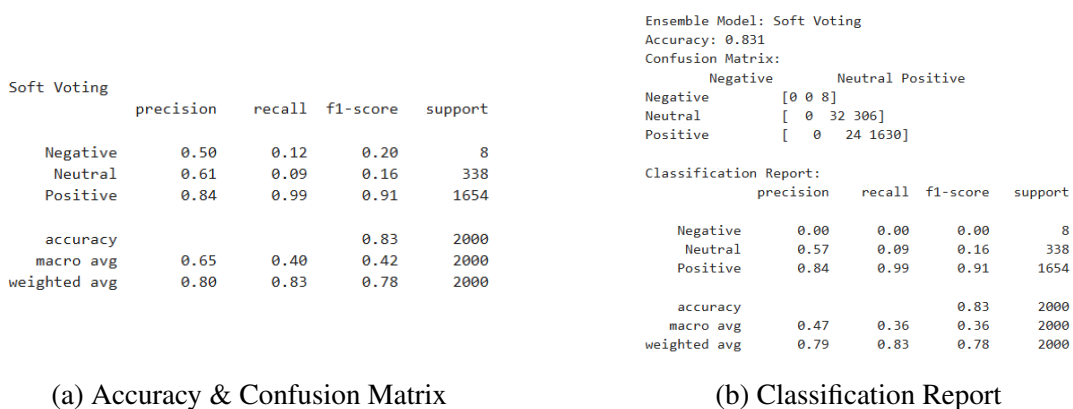


Figure 7.11: Soft Voting Ensemble

	Model	Accuracy	Precision	Recall	F1-score
0	Linear SVC	0.8320	0.796776	0.8320	0.797825
1	SVC	0.8320	0.797183	0.8320	0.772467
2	Logistic Regression	0.8275	0.775926	0.8275	0.757121
3	SGD Classifier	0.8315	0.795976	0.8315	0.770678
4	Decision Tree	0.8020	0.782270	0.8020	0.790491
5	K Nearest Neighbors	0.8055	0.740364	0.8055	0.760608
6	Multinomial Naive Bayes	0.8310	0.814527	0.8310	0.761623
7	Random Forest	0.8400	0.812360	0.8400	0.809479
8	Gradient Boosting	0.8260	0.782517	0.8260	0.752513

Figure 7.12: Overall Performance Comparison of All 9 Models

```

tensor([[ 0.4074, -0.1831, -0.7493],
        [ 0.2452,  0.0394, -0.4589],
        [ 0.2538, -0.1134, -0.5938],
        [ 0.3032, -0.1011, -0.4310],
        [ 0.1403,  0.0381, -0.3535],
        [ 0.1870,  0.0238, -0.4031],
        [ 0.3372, -0.1502, -0.6944],
        [ 0.2856,  0.0097, -0.5953]], grad_fn=<AddmmBackward0>)

```

[8, 3] shows the size of the batch and the number of classes the data is shown as [negative, neutral, positive]

Figure 7.13: BERT Model Output

predicted_sentiment	Negative	Neutral	Positive
Company name			
1 India Family Mart	0	0	1
1K Networks	0	0	1
1point1 Solutions	0	1	0
20 Microns	0	0	1
20Cube Logistics	0	0	1
...	...	...	...
we-do-IT	0	0	1
yellow.ai	0	1	0
Škoda Auto India	0	0	1
Škoda Auto Volkswagen	0	0	1
“Bharat Financial Inclusion Limited”	0	0	1

9100 rows × 3 columns

Figure 7.14: Company Wise Sentiment

predicted_sentiment	Negative	Neutral	Positive
Sector			
1 Lakh+ Employees	0	0	1
1-10 Employees	0	0	10
10k-50k Employees	0	2	18
10k-50k Employees (Global)	0	0	3
11-50 Employees	0	0	16
...	...	...	...
Travel & Tourism	0	0	43
Urban Transport	0	0	25
Visakhapatnam +8 more	0	0	1
Waste Management	0	0	24
bangalore +11 more	0	0	1

222 rows × 3 columns

Figure 7.15: Sector Wise Sentiment

predicted_sentiment	Negative	Neutral	Positive
Company location			
\tMoon Township +87 more	0	0	1
Seoul +12 more	0	0	2
Aarhus +38 more	0	0	1
Abu Dhabi +10 more	0	0	1
Abu Dhabi +11 more	0	0	1
...	...	...	...
Zurich +3 more	0	0	1
Zurich +88 more	0	0	1
chennai +31 more	0	0	1
shenzhen +43 more	0	0	1
Pittsburgh +10 more	0	0	1

1696 rows × 3 columns

Figure 7.16: Location Wise Sentiment

CHAPTER 8

CONCLUSION AND FURTHER ENHANCEMENTS

8.1 CONCLUSION

The project was executed as proposed for an intelligent feedback system for enhancing placement experiences. This was done through evaluating the employee dataset obtained from AmbitionBox and using nine machine learning models and natural language processing. The project was successful in processing the large volume of data that it was fed and did so with the highest accuracy of 84.

The dataset from AmbitionBox was first loaded into the Python code and was pre processed so that it is good to be used to train the nine machine learning models. The data was cleaned by using TF-IDF feature extraction to extract the features from the dataset. Once this was done, the various machine learning models were trained, namely, they were linear SVC, SVC, logistic regression, SGD classifier, decision tree, k nearest neighbours, multinomial naive bayes, random forest and gradient boosting. In addition to this, ensemble learning techniques such as hard voting and soft voting were made use of to improve predictions and stability.

Among the nine different machine learning models that were used, we obtained the highest accuracy from the random forest classifier with 84. Along with this, from evaluating the results, we can say that the class imbalance with the presence of fewer negative classes did affect the prediction of this particular sentiment class.

In order to be more transparent and more understandable to the end user, explainable AI techniques such as SHAP and LIME have also been incorporated. SHAP makes use of concepts from game theory and assigns weights or values to each feature. The feature with the highest weight is considered to be more important. LIME is used to give local explanations, and values are assigned to the features of the dataset. If the values are more positive are pushed towards the predicted class, and the negative values are pushed away from the predicted class.

Finally, aggregated sentiment analysis has also been incorporated across companies, sectors, and locations. This enables the user to view sentiment analysis for these particular areas. This is done to make useful data driven decision making.

All in all, the project showcases the effectiveness of using nine machine learning models along with learning approaches with explainable AI techniques to give a reliable sentiment analysis system.

8.2 FUTURE ENHANCEMENTS

Additional features can be integrated to improve its usability in the future.

- Real Time Data integration
- Multilingual Sentiment Analysis

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Appendix A- Sample Code

Python Programming

```
Importing the dataset into our program
df = pd.read_csv("AmbitionBoxDataset.csv")

Using stop words and setting instance for lemmatizer
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

Cleaning the data present in the review section
df['clean_review'] = df['Reviews'].apply(clean_text)

Unique ratings are retrieved to set the parameters for the dataset
df['Rating'].unique()

Parameters are now set for the dataset we are using
def rating_to_sentiment(rating):
    if rating <= 2.5:
        return "Negative"
    elif rating <= 3.5:
        return "Neutral"
    else:
        return "Positive"

Checking how many positive, negative and neutral reviews there
df['sentiment'].value_counts()

Printing the shape of the feature matrix and the number of features present in the dataset
print(X.shape)
print(len(tfidf.get_feature_names_out()))

Shape of the X_train data and X_test data are obtained
print(X_train.shape)
print(X_test.shape)

Models are specified that will be used to train the model
models = {
    "Linear SVC": LinearSVC(),
    "SVC": SVC(kernel='rbf', probability=True),
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "SGD Classifier": SGDClassifier(random_state=42),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "K Nearest Neighbors": KNeighborsClassifier(n_neighbors=5),
    "Multinomial Naive Bayes": MultinomialNB(),
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),
    "Gradient Boosting": GradientBoostingClassifier(random_state=42),
```

```
}
```

Accuracy and Confusion Matrix are obtained for all the nine models

```
results = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
    cm = confusion_matrix(y_test, y_pred)
    results[name] = {
        "accuracy": acc,
        "confusion_matrix": cm
    }

print("*" * 100)
print(f"Model: {name}")
print(f"Accuracy: {acc:.4f}")
print("Confusion Matrix:")
print(cm)
```

Now we obtain the classification report for the 9 ML models that we have used

Hard Voting

```
for name, model in models.items():
    y_pred = model.predict(X_test)
    print("*" * 100)
    print(f"Model: {name}")
    print(classification_report(y_test, y_pred, zero_division=0))

Voting classifier is used to combine and use all the nine ML models
voting_hard = VotingClassifier(
    estimators=[
        ("lsvc", models["Linear SVC"]),
        ("svc", models["SVC"]),
        ("lr", models["Logistic Regression"]),
        ("sgd", models["SGD Classifier"]),
        ("dt", models["Decision Tree"]),
        ("knn", models["K Nearest Neighbors"]),
        ("nb", models["Multinomial Naive Bayes"]),
        ("rf", models["Random Forest"]),
        ("gb", models["Gradient Boosting"])
    ],
    voting="hard"
)

voting_hard.fit(X_train, y_train)
y_pred_hard = voting_hard.predict(X_test)
```

Soft Voting

```
voting_soft = VotingClassifier(
    estimators=[
        ("svc", models["SVC"]),
```

```

        ("lr", models["Logistic Regression"]),
        ("dt", models["Decision Tree"]),
        ("knn", models["K Nearest Neighbors"]),
        ("nb", models["Multinomial Naive Bayes"]),
        ("rf", models["Random Forest"]),
        ("gb", models["Gradient Boosting"])
    ],
    voting="soft"
)
voting_soft.fit(X_train, y_train)
y_pred_soft = voting_soft.predict(X_test)

Getting the classification report for both hard and soft voting methods
print("Hard Voting")
print(classification_report(y_test, y_pred_hard, zero_division=0))
print("Soft Voting")
print(classification_report(y_test, y_pred_soft, zero_division=0))

Constructing the comparison table in order to compare all the models on various performance
comparison = []
for name, model in models.items():
    y_pred = model.predict(X_test)
    comparison.append([
        name,
        accuracy_score(y_test, y_pred),
        precision_score(y_test, y_pred, average='weighted', zero_division=0),
        recall_score(y_test, y_pred, average='weighted', zero_division=0),
        f1_score(y_test, y_pred, average='weighted', zero_division=0)
    ])
comparison_df = pd.DataFrame(
    comparison,
    columns=["Model", "Accuracy", "Precision", "Recall", "F1-score"]
)
comparison_df

Converting numerical values to positive, negative and neutral
df['sentiment'] = df['sentiment'].map({
    'negative': 0,
    'neutral': 1,
    'positive': 2
})

Evaluates the model based from train_loader and shows the size of the batch and the number o
model.eval()
for batch in train_loader:
    outputs = model(
        input_ids=batch['input_ids'],
        attention_mask=batch['attention_mask']
    )

```

```

print(outputs.logits)
break

```

Using SHAP to obtain weights of each feature. A feature with more weight has more importance

```

import torch
model.eval()
for batch in train_loader:
    with torch.no_grad():
        outputs = model(
            input_ids=batch['input_ids'],
            attention_mask=batch['attention_mask'],
            output_attentions=True
        )
    logits = outputs.logits
    attentions = outputs.attentions
    #prediction
    preds = torch.argmax(logits, dim=1)
    print("Predictions:", preds)
    tokens = tokenizer.convert_ids_to_tokens(batch['input_ids'][0])
    attention = attentions[-1][0].mean(dim=0)
    #importance of each word
    importance = attention[0]
    print("\nWord Importance:\n")
    for word, score in zip(tokens, importance):
        print(f"{word} : {float(score):.4f}")
    break

```

SHAP gives features positive and negative values. Where positive values push towards the pre

```

import torch
import numpy as np
from lime.lime_text import LimeTextExplainer
class_names = ["Negative", "Neutral", "Positive"]
explainer = LimeTextExplainer(class_names=class_names)
def predict_proba(texts):
    inputs = tokenizer(
        texts,
        return_tensors="pt",
        padding=True,
        truncation=True,
        max_length=128
    )
    with torch.no_grad():
        outputs = model(
            input_ids=inputs["input_ids"],
            attention_mask=inputs["attention_mask"]
        )
    probs = torch.softmax(outputs.logits, dim=1)
    return probs.numpy()
#sample text

```

```

sample_text = df['clean_review'].iloc[0]
#explain prediction
exp = explainer.explain_instance(
sample_text,
predict_proba,
num_features=10
)
#explanation
print("\nLIME Explanation:\n")
for feature, weight in exp.as_list():
print(f"{feature} : {weight:.4f}")

Give us an independent sentiment analysis of the company, sector and location
company_sentiment = df_analysis.groupby(
["Company name", "predicted_sentiment"]
).size().unstack(fill_value=0)
sector_sentiment = df_analysis.groupby(
["Sector", "predicted_sentiment"]
).size().unstack(fill_value=0)
location_sentiment = df_analysis.groupby(
["Company location", "predicted_sentiment"]
).size().unstack(fill_value=0)

company_sentiment
sector_sentiment
location_sentiment

Saves the company, sector, and location sentiment DataFrames as CSV files
company_sentiment.to_csv("company_sentiment.csv")
sector_sentiment.to_csv("sector_sentiment.csv")
location_sentiment.to_csv("location_sentiment.csv")

```

Appendix B- Publication details

- 2026 Annual Interdisciplinary Student Research Conference (AUI-AISRC).

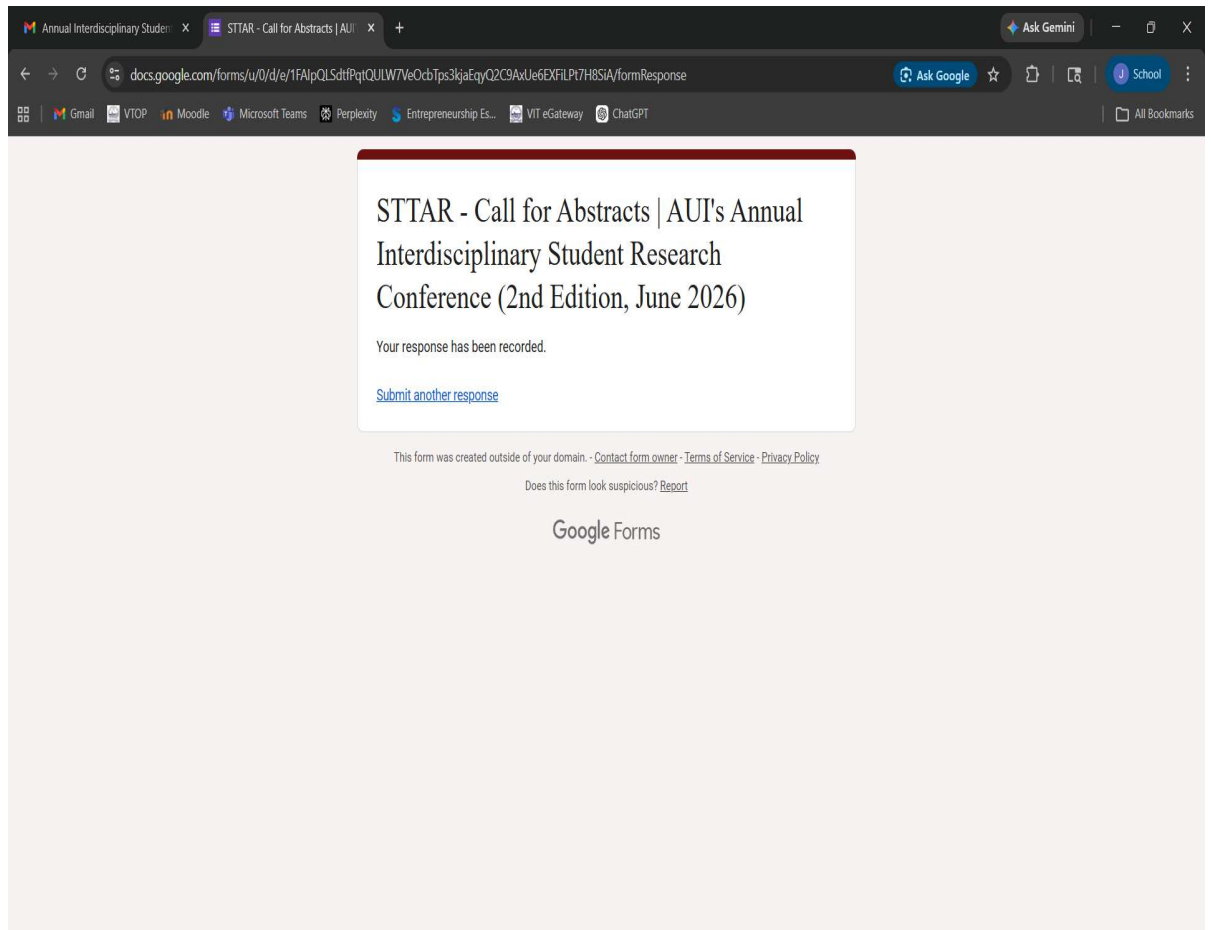


Figure 8.1: Paper Abstract Submission